

Characterisation of RFI contaminations for single dish observations with the MeerKAT telescope

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12 July 2025

Submitted in partial fulfillment of a structured masters degree at AIMS South Africa



Abstract

The MeerKAT Large Area Synoptic Survey (MeerKLASS) is one of the largest initiatives conducted with the MeerKAT telescope, a precursor to the SKA observatory in South Africa. Its aim is to measure the large-scale distribution of neutral hydrogen in order to study the evolution of dark matter and dark energy using the autocorrelation data from MeerKAT's single-dish observations. A major challenge in radioc cosmology is Radio Frequency Interference (RFI), which can compromise data analysis and needs to be carefully flagged. This project explores a novel approach to analysing RFI flags to gain a more comprehensive understanding of RFI contamination. We begin with a detailed examination of flags over time and frequency for different blocks of observations and then investigate the relationship between RFI flag patterns and antenna positions. By defining a dissimilarity metric between the flagging patterns of antenna pairs, we find a positive correlation between RFI similarity and physical proximity. Applying a manifold learning algorithm to these dissimilarities successfully recovers a two-dimensional map that reflects the known layout of the MeerKAT antennas. The technique also proves to be a powerful diagnostic tool, autonomously identifying outlier antennas affected by localized interference. This work validates a new method for RFI characterization, reframing interference patterns as a useful probe of instrumental and environmental systematics affecting cosmological radio surveys.

Keywords: MeerKAT, MeerKLASS, Neutral hydrogen, RFI, Manifold Learning

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Mushtaha Mutasim Mohammed El-Amin, 12 June 2025

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1. Introduction

Mapping the large-scale structure (LSS) of the universe is a cornerstone of modern cosmology. While optical and infrared galaxy surveys have made significant progress in tracing the cosmic web, alternative techniques such as HI intensity mapping offer a complementary and efficient approach to probe the same underlying matter distribution. Rather than resolving individual galaxies, intensity mapping integrates the unresolved 21 cm emission from neutral hydrogen HI across relatively large angular pixels, enabling high-speed surveys over cosmological volumes (Wang et al., 2021; Cunnington et al., 2023).

The 21 cm hyperfine line of HI redshifted by cosmic expansion allows us to construct a three-dimensional map of hydrogen distribution in the universe. This line traces the locations of neutral hydrogen gas, which resides predominantly in dark matter halos after the Epoch of Reionization ($z < 6$), and thus acts as a biased tracer of matter. HI intensity mapping is especially promising at intermediate redshifts ($z \sim 0.1\text{--}1.5$), a range where optical surveys become increasingly resource-intensive (Wang et al., 2021).

The MeerKAT telescope, located in the Karoo region of South Africa (Figure 1.1), is a 64-dish radio interferometer that functions as a key precursor to the SKA Observatory (SKAO), an international project to build the world's largest and most sensitive radio telescope, designed to explore the early universe, test theories of gravity, and map cosmic magnetism with unprecedented resolution and survey speed. With its high sensitivity and flexible observing modes, MeerKAT can perform HI intensity mapping both as an interferometer and in single-dish (autocorrelation) mode. The MeerKAT Large Area Synoptic Survey (MeerKLASS) is a flagship project designed to exploit this capability, observing wide fields of the southern sky in the L-band and UHF-band (Collaboration et al., 2025; Cunnington et al., 2023). The recent L-band deep-field survey scanned a 236 deg^2 region with repeated coverage, yielding one of the deepest single-dish intensity maps to date.



Figure 1.1: An overview of the MeerKAT radio telescope in the Karoo in the Northern Cape, South Africa. Image taken from the official website of the South African Radio Astronomy Observatory (SARAO)

Despite the promise of 21 cm cosmology, one of the key challenges lies in the mitigation of radio frequency interference (RFI) spurious signals generated by human-made sources such as satellites, communication towers, or electronic equipment. Unlike diffuse galactic foregrounds, RFI can vary rapidly in time and

space, and often appears as transient, high-amplitude spikes in the data (Engelbrecht et al., 2024). If unaddressed, RFI contaminates the measured brightness temperature fluctuations and biases the inferred cosmological signal.

To address this, sophisticated flagging strategies are employed as part of the calibration pipeline. The MeerKLASS processing chain includes multiple RFI detection stages, including per-channel outlier removal, iterative map cleaning, and strong-flagging heuristics. These procedures are implemented in tools such as AOFlogger and extended with self-calibration routines to minimize RFI leakage and recover accurate HI maps (Collaboration et al., 2025; Wang et al., 2021). Moreover, recent studies show that analyzing the spatial and temporal distribution of flagged data may reveal instrumental systematics or environmental signatures, such as antenna-specific contamination or satellite crossings (Engelbrecht et al., 2024).

For HI intensity mapping studies it is important to properly flag and mask RFI. The distribution of these flags is, however, itself an observational product: each flag records where, when, and from which antenna the system has detected contamination. If particular RFI sources, such as satellites, ground transmitters, or electronic couplers, interact differently with particular antennas or sky regions, studying the flag statistics should help better understand the behavior of the telescope. Modern dimensionality reduction techniques (e.g. MDS, LLE) are well suited to reveal low-dimensional structures in such high-dimensional data. We therefore investigate whether the vectors of the flag fraction in the 64 dishes of MeerKAT contain recoverable spatial information about the array.

The remainder of this report is structured as follows. Chapter 2 provides the essential scientific and technical background. It starts by describing in a bit more detail intensity mapping and thus the context in which the data analyzed in this thesis have been collected and their scientific purpose. It then describes how RFI is flagged using standard techniques and presents the methodology of Manifold Learning. Chapter 3 presents the MeerKAT survey data used in this study in detail and provides an initial description of their statistical properties. Chapter 4 outlines the methodology we used and reports and discusses the results. Finally, Chapter 5 summarizes the key findings and suggests potential directions for future research.

2. Background

In this chapter we outline the essential scientific and technical background for this research. We begin with an introduction to the principles of neutral hydrogen intensity mapping and the fundamental challenge posed by radio frequency interference (RFI). We then describe the AOFlagger tool used for RFI flagging and conclude with an overview of the manifold learning framework that forms the basis of our analysis.

2.1 Neutral Hydrogen and Intensity Mapping

Neutral hydrogen sometimes referred to as atomic hydrogen or HI is one of the various forms of hydrogen, the most prevalent element in the universe. The simplest hydrogen atom is called HI. It is electrically neutral because it only has one proton and one electron, which cancel each other's charge. The quantum structure of this neutral hydrogen is unique. Depending on whether its spin is parallel or antiparallel to that of the proton, its electron can exist in two slightly different energy states. The antiparallel alignment represents a slightly lower energy state.

When the electron spontaneously transitions from the higher (parallel spin) to the lower (antiparallel spin) energy state, it releases a photon with a specific energy of approximately 5.87 micro-electron volts (μeV). This emitted radiation is referred to as the 21 cm emission line because this energy is equivalent to a frequency of approximately 1420 MHz and a wavelength of approximately 21 cm. With an average lifetime of roughly 10 million years for the excited state, this energy change is extremely rare. However, the enormous amount of neutral hydrogen in the Universe guarantees that this redshifted 21cm radiation enables HI to serve as a biased tracer of cosmic structure formation, offering insights into the evolution of matter distribution across cosmological timescales (Cunnington, 2019).

Traditional galaxy surveys rely on the detection and spectroscopic confirmation of individual galactic sources, a methodology that becomes increasingly challenging at intermediate redshifts ($z \sim 0.4$) due to source confusion, flux limitations, and observational biases. An alternative approach to mapping the Universe's LSS is to use the intensity mapping method to map the 21 cm emission from cosmic neutral hydrogen. The method involves low angular resolution surveys measuring the collective emission from neutral hydrogen aggregations and many unresolved galaxies.

The method is shown in Figure 2.1. The spatial distribution of galaxies, each with a distinct HI mass, is shown in the left panel. A radio telescope surveying this sky area would capture the distribution displayed in the right panel. The observed variations in this field trace the underlying matter density field by directly matching variations in the HI/galaxy distribution.

Observations of intensity mapping across cosmic time are made possible by the phenomenon of cosmological redshift. The wavelength of the 21cm emission from neutral hydrogen stretches in proportion to the cosmic scale factor as it moves through the expanding Universe. Through this process, lower observed frequencies replace the 1420 MHz rest-frame frequency. The relationship between redshift and observed frequency, expressed as

$$z = \frac{\nu_{\text{emitted}} - \nu_{\text{observed}}}{\nu_{\text{observed}}},$$

where z denotes the redshift, ν_{emitted} is the rest-frame (emitted) frequency of the signal, and ν_{observed} is the frequency at which the signal is detected by the observer. This expression establishes a direct correlation between the observational frequency and the corresponding cosmic epoch, facilitating tomographic

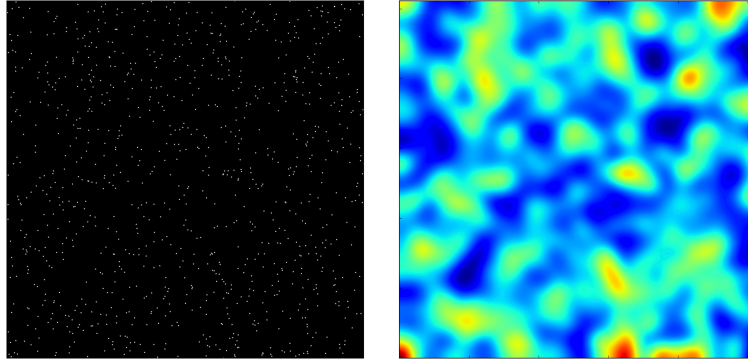


Figure 2.1: Illustration of the intensity mapping method (image credit: [Villaescusa-Navarro](#)).

mapping of the Universe's evolution.

2.2 Radio Frequency Interference (RFI)

RFI refers to unwanted electromagnetic signals from man-made sources that contaminate radio astronomical observations. These sources include terrestrial communication systems, aircraft radar, satellite constellations and consumer electronics. In observational data, RFI typically appear as narrow or broadband features with high amplitude in the time/frequency domain. They are often intermittent, non-stationary and vary greatly depending on the antenna and direction of pointing.

RFI is particularly problematic in HI intensity mapping surveys, where the desired 21 cm signal from neutral hydrogen is extremely faint on the order of a few millikelvins. Even brief or localised interference can significantly distort statistical measurements such as the power spectrum of the brightness temperature. Residual RFI after calibration may also distort foreground subtraction, distort gain solutions and introduce unwanted correlations across frequencies or antennas ([Wang et al., 2021](#)).

Although the MeerKAT telescope is located in the radio quiet Karoo region of South Africa, it is still susceptible to RFI, particularly from satellite transmissions and nearby cell towers. For example, frequencies around 1015–1025 MHz often coincide with emissions from Global Navigation Satellite Systems (GNSS), such as GPS. Depending on the satellite position and observation geometry, these emissions affect certain time blocks and subsets of the array ([Engelbrecht et al., 2024](#)). The growing number of satellite constellations in low Earth orbit (LEO) further exacerbates this problem and has become a significant concern for future deep field intensity mapping campaigns. Given the sensitivity of HI experiments to contamination, effective and automatic RFI flagging is essential.

2.3 RFI Flagging with AOFlagger

RFI Flagging refers to the process of identifying and masking corrupted data samples in time/frequency visibilities prior to further calibration and analysis. This is to ensure that downstream data products

such as sky maps and power spectra are only derived from clean, uncontaminated measurements.

In the MeerKLASS survey, RFI is handled using AOFlagger, an automated flagging tool that recognises and removes corrupted data segments from time/frequency visibilities ([Offringa](#)).

AOFlagger combines statistical techniques with morphological filtering to identify RFI signatures in a robust and scalable way. It is mainly based on two components:

- **SumThreshold algorithm:** a multi-scale detection method that flags groups of samples if their combined power exceeds a threshold.
- **Scale-Invariant Rank (SIR) operator:** a morphological filter that cleans up fragmented regions and improves sensitivity to weak or diffuse RFI.

AOFlagger operates on 2D arrays of visibilities (time $\times$ frequency) and generates a binary flag mask indicating which samples should be excluded from scientific analysis.

In the MeerKLASS processing pipeline, AOFlagger is applied in several stages. The first flagging removes RFI at high power prior to bandpass and flux calibration. After self-calibration, a second round removes the remaining interference, which becomes more visible after the removal of sky and instrument models. The resulting flags are used to:

- Exclude corrupted voxels during map generation and stacking.
- calculate the RFI statistics per dish and per scan.
- Correlate flagging rates with antenna position, sky coordinates and observation time.

2.4 Manifold Learning

High-dimensional datasets are prevalent in modern scientific research, posing challenges for visualisation, interpretation, and analysis. To address this, a variety of dimensionality reduction techniques have been developed that aim to reveal meaningful low-dimensional structures embedded within high-dimensional data. Among these, manifold learning methods provide powerful tools for discovering the underlying geometry of datasets by preserving pairwise relationships or neighbourhood structures in a lower-dimensional embedding. These techniques are especially valuable when the data is hypothesised to lie on or near a low-dimensional manifold, as they enable more interpretable and structure-preserving representations of complex data ([Izenman, 2008](#)).

2.4.1 The Manifold Hypothesis

The foundation of manifold learning rests on the manifold hypothesis, which states that high-dimensional data encountered in real-world applications often lie on or near a low-dimensional manifold embedded within the higher-dimensional ambient space. This means that although the data may be represented using many observed variables or features, their intrinsic structure can be described using a much smaller number of underlying parameters ([Fefferman et al., 2016](#)).

Essentially, the manifold hypothesis suggests that the true degrees of freedom that determine the variability in the data are significantly fewer than the number of dimensions in which the data are measured. As a result, the observed high-dimensional data occupy only a thin, curved subspace, or manifold, within the larger space ([Meilă and Zhang, 2024](#)). In the context of radio interferometry, antenna flagging patterns in time-frequency space represent high-dimensional binary data that may encode low-dimensional spatial relationships between the antennas.

2.4.2 Overview of Manifold Learning Techniques

Manifold learning includes a suite of algorithms designed to capture the underlying structure of data, such as:

Multidimensional Scaling (MDS): Seeks a low-dimensional embedding that preserves the pairwise distances between data points. MDS provides a linear embedding and forms the basis for more advanced nonlinear methods such as Isomap (Torgerson, 1952).

Locally Linear Embedding (LLE): Focuses on preserving local neighborhood relationships by reconstructing each data point from its nearest neighbors and seeking a low-dimensional embedding that maintains these relationships Roweis and Saul (2000).

Isometric Mapping (Isomap): Extends classical MDS by preserving geodesic distances computed over a neighborhood graph, effectively capturing global geometric structures Tenenbaum et al. (2000).

Figure 2.2 shows a visual comparison of different algorithms applied to a classic “Swiss Roll” dataset. This visualisation shows how the different methods attempt to “unroll” the 3D data into a 2D plane while preserving the intrinsic structure. This makes it clear how each technique captures the underlying diversity with varying degrees of success.

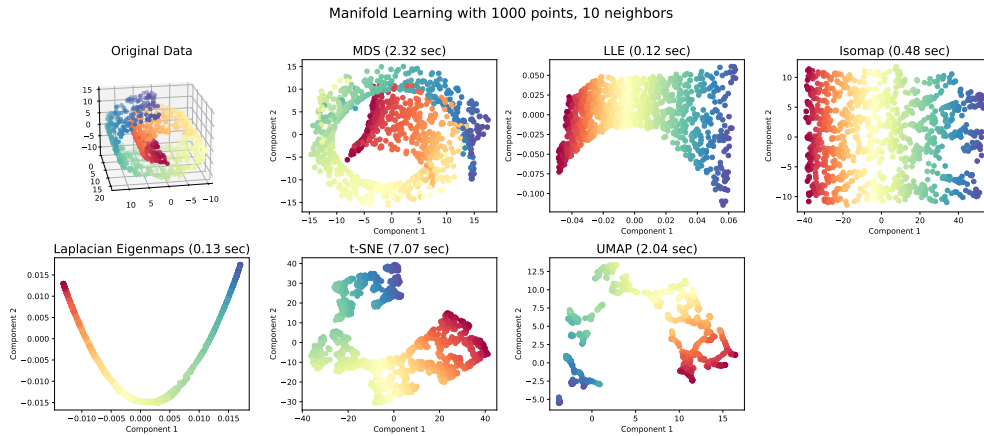


Figure 2.2: A comparison of manifold learning algorithms applied to the synthetic 'Swiss Roll' dataset.

While each method offers unique advantages, MDS is an ideal framework for this particular analysis. The main reason for this is that our method for comparing antennas, which is calculating the Hamming distance between their RFI flagging patterns as we will show in chapter 4, inherently produces a pairwise dissimilarity matrix. MDS is specifically designed to use such a matrix as a direct input, so that more complex feature vectors are not required. The aim of this work is therefore to apply MDS to this RFI-derived dissimilarity matrix to create a 2D map of the antennas. The main objective is to determine whether the configuration of the antennas on this map reflects the known physical layout of the MeerKAT array, thus testing the hypothesis that the RFI flag patterns encode the geometry of the telescope.

3. The MeerKLASS survey data and their RFI environment

The survey data used in this study is taken from the MeerKLASS observational campaign, conducted using the MeerKAT radio telescope array consisting of 64 dish or antennas. The 64 antennas positions are specified in the local MeerKAT coordinate system, which includes three spatial dimensions: altitude, referring to the local elevation relative to the array centre; longitude, representing the local azimuthal (east–west) position; and elevation, indicating the height above a local reference level. These coordinates define the configuration of the antennas within the array. Figures 3.1 and 3.2 show the layout of MeerKAT’s antennas, consisting of a dense core and spiral arms.

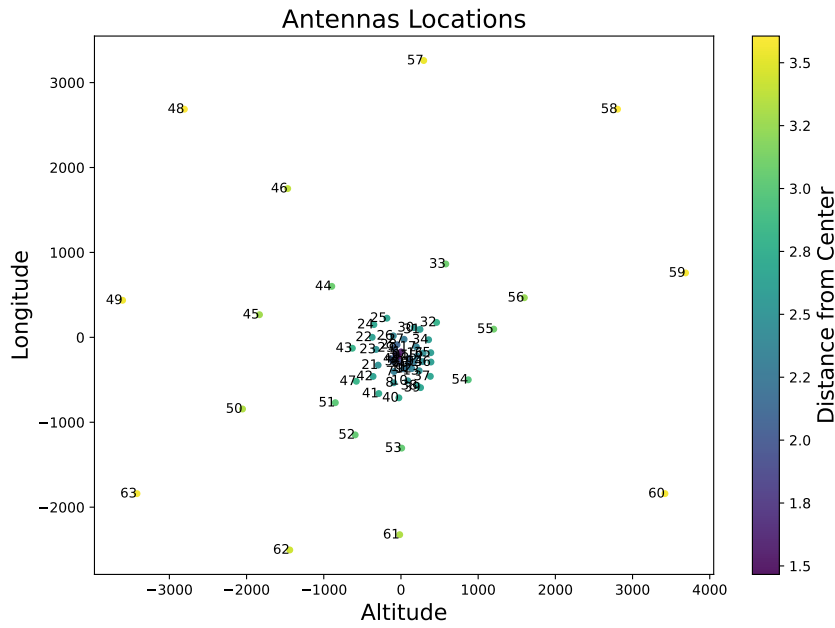


Figure 3.1: Complete spatial layout of all 64 MeerKAT antennas plotted in altitude-longitude coordinates. The color scale represents log-scaled distances from the central reference antenna (antenna 0).

3.1 MeerKLASS RFI flags description

We analysed a subset of the L-band MeerKLASS survey data consisting of five separate observation blocks, each pointed at the same patch in the sky. The data were gathered between September and December 2021. Each block contained three-dimensional auto-correlation visibilities as a function of frequency and time. The pointing of the antennas changes every 2 seconds as a function of time. The blocks contained 4096 distinct channels, giving a frequency coverage ranging from 856 to 1712 MHz. This range corresponds to HI emission from redshifts $z \approx 0.2-0.8$.

Each observational block as shown in Table 3.1 represents a complete observation sequence containing both scan and track modes.

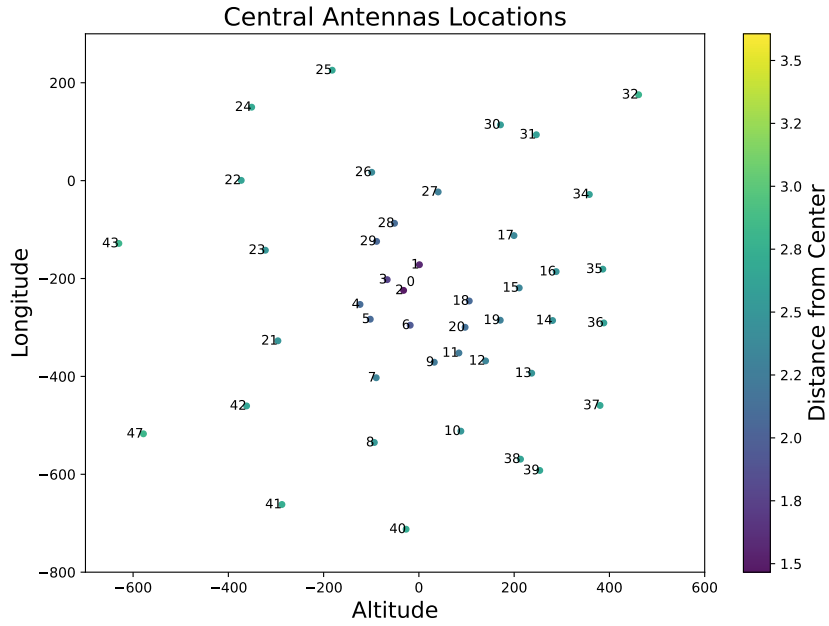


Figure 3.2: Detailed view of the central core region of the MeerKAT array, showing the dense concentration of antennas near the reference antenna (antenna 0). The colour scale represents log-scaled radial distances from the array centre.

Block	Date	Antennas
Block 1 (1640712986)	2021-12-28 17:36:26	62
Block 2 (1640799689)	2021-12-29 17:41:29	60
Block 3 (1638639082)	2021-12-04 17:31:22	56
Block 4 (1639157507)	2021-12-10 17:31:47	48
Block 5 (1631552188)	2021-09-13 16:56:28	55

Table 3.1: Summary of observation blocks, their timestamps, and active antenna count.

For the work in this thesis, we focus on flags obtained with the AOFlagger only. Each of the analysed blocks contains:

- Flags arrays: Binary matrices of shape $(n_antennas, n_timesteps, n_frequencies)$ indicating RFI-contaminated data points. Figure 3.3 shows how one antenna can be visualised.
- Frequency information: L-band coverage (856–1712 MHz) with 0.2 MHz resolution.
- Observation metadata: Contains timestamp, azimuth and elevation coordinates (local horizontal coordinate system with respect to the telescope location), right ascension and declination (equatorial coordinate system with respect to the celestial sphere), and scan/track identifiers that define the two-stage MeerKLASS observation strategy. Track mode (~ 8 minutes), targets bright calibration sources for bandpass calibration and flux scaling, while in scan mode (~ 100 minutes), a constant elevation scan is performed with dishes moving in azimuth, as shown in Figure 3.4.

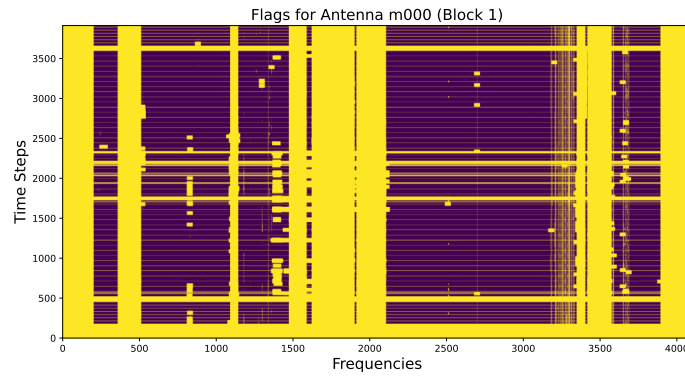


Figure 3.3: Binary RFI flagging map for antenna m000 in Block 1, showing time on the vertical axis and frequency on the horizontal axis. Yellow indicates flagged (corrupted) samples, and purple indicates unflagged (clean) data.

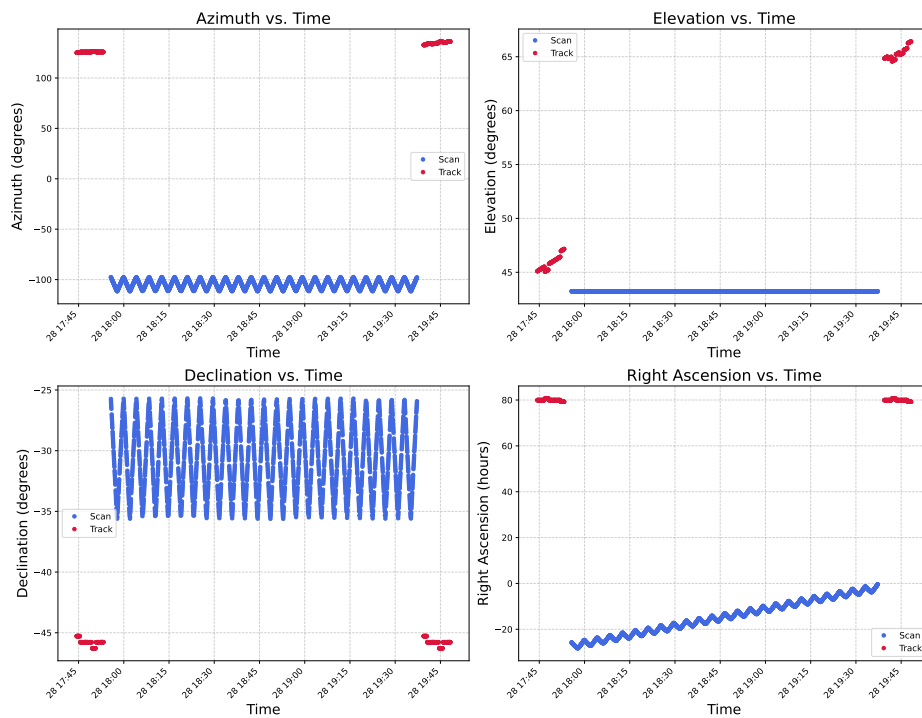


Figure 3.4: Telescope sky pointing coordinates demonstrating MeerKLASS's two phase observational strategy. Track mode shows rapid movement targeting calibrator sources, while scan mode maintains constant elevation with azimuthal drift scanning to map the sky.

Prior to analysis, antennas that were heavily flagged during specific observation blocks, such as m052, m054, m060, and others identified per block, were excluded as shown in Figure 3.5.

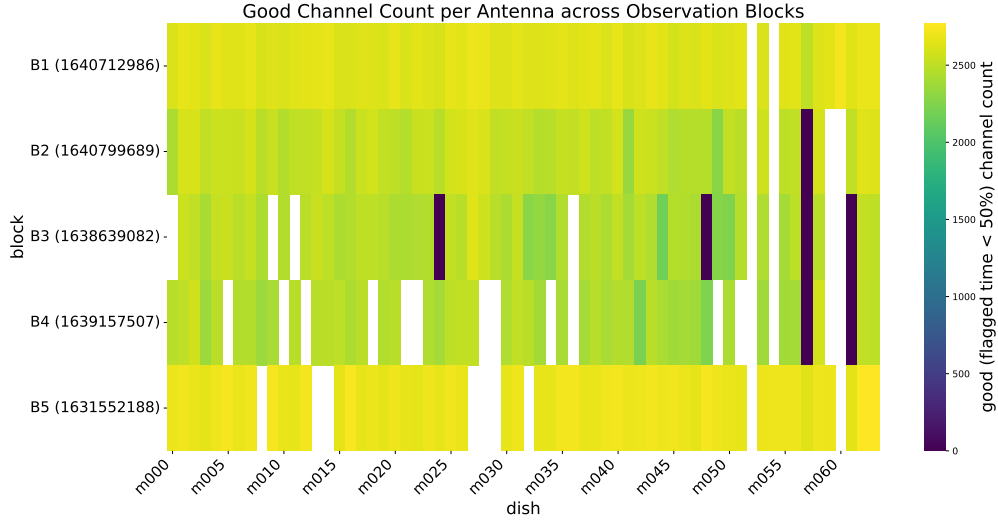


Figure 3.5: Data quality assessment showing the count of good frequency channels (flagged time < 50%) per antenna across five observation blocks. Yellow regions indicate high-quality data with most channels available, while dark blue/purple regions show heavily flagged antennas with few usable channels. White areas represent antennas that were completely excluded from specific blocks due to poor performance.

3.2 Exploratory Data Analysis

We conduct a comprehensive analysis of flagging patterns across the L-band frequency range to identify persistent sources of interference and to characterise their spectral signatures over time. This is achieved by computing two statistical summaries from the 3D flagging data cube, denoted as $F(a, t, f)$, where:

- a indexes antennas.
- t indexes time samples,
- f indexes frequency channels,

Average Flagging Rate per Frequency

The average flagging rate per frequency channel, as defined in Equation 3.2.1, quantifies the fraction of time antenna samples flagged for each frequency channel:

$$\text{Average Flagging Rate}(f) = \frac{1}{T \cdot A} \sum_{t=1}^T \sum_{a=1}^A F(a, t, f) \quad (3.2.1)$$

where T is the total number of time samples, and A is the number of active antennas in a given block.

Average Flagging Rate per Time Sample

To capture the temporal evolution of RFI, we computed the average flagging rate per time sample, as defined in Equation 3.2.2

$$\text{Average Flagging Rate}(t) = \frac{1}{F \cdot A} \sum_{f=1}^F \sum_{a=1}^A F(a, t, f) \quad (3.2.2)$$

This metric describes how the total flagging level changes over time, averaged across frequency and antenna dimensions. These two metrics are visualised in Figure 3.6 for all five observation blocks and provide a detailed characterisation of the RFI environment.

The average flagging rate per time sample, shown in the left panels, reveals the temporal structure of the interference and highlights significant variations in the total RFI power both between and within observations. These plots clearly show the different RFI characteristics of the scan and track observation modes. Conversely, the average flagging rate per frequency, plotted in the right panels, highlights the persistent spectral features. The most noticeable feature in these plots is the sharp peaks that consistently appear at the same frequencies in different observations. This pattern indicates that the interference comes from fixed sources, e.g. local transmitters.

3.3 Satellites and cellphone tower RFI

To recognize persistent RFI sources, we analyze the flagged data for evidence of known transmitters such as satellites and cell towers. The average flagging rate, shown in Figure 3.6, reveals specific spectral bands that are consistently contaminated. These bands often correspond to known communication frequencies such as the Global Navigation Satellite Systems (GNSS).

A specific example of satellite detection is shown in Figure 3.7 for block 1. This detection is achieved computationally by first calculating the average RFI flagging fraction for each time sample during the scanning by averaging the binary flag data across all active antennas and the full frequency range. The time of maximum interference is then determined in order to find the index of the time step that has the highest value in this average flagging array. Finally, we use this timestep index to pick a frequency from the list of frequency channels and label the result as the frequency of the satellite. The analysis shows a distinct peak in RFI at approximately 1067 MHz when the telescope is pointed at a specific region of the sky, which is a clear indication of an orbiting satellite crossing the telescope's field of view.

The localized source of interference is identified by plotting the flagged data fraction as a function of the telescope's sky position, in particular right ascension (RA) and declination (DEC), as shown in Figure 3.7.

3.4 Sky-dependent RFI patterns

The spatial distribution of RFI is analyzed by visualizing flagging patterns as a function of the telescope pointing direction. The correlation between the RFI flagging fraction and the telescope pointing coordinates, right ascension (RA), declination (DEC), azimuth (AZ) and elevation (EL), is evaluated for each observation block, as shown in Figure 3.8. This analysis can be used to determine whether RFI depends on the direction of observation. For example, blocks 2, 3, and 4 show notable correlations between flagging and the right ascension, which indicates a directional dependence of the interference in these observations.

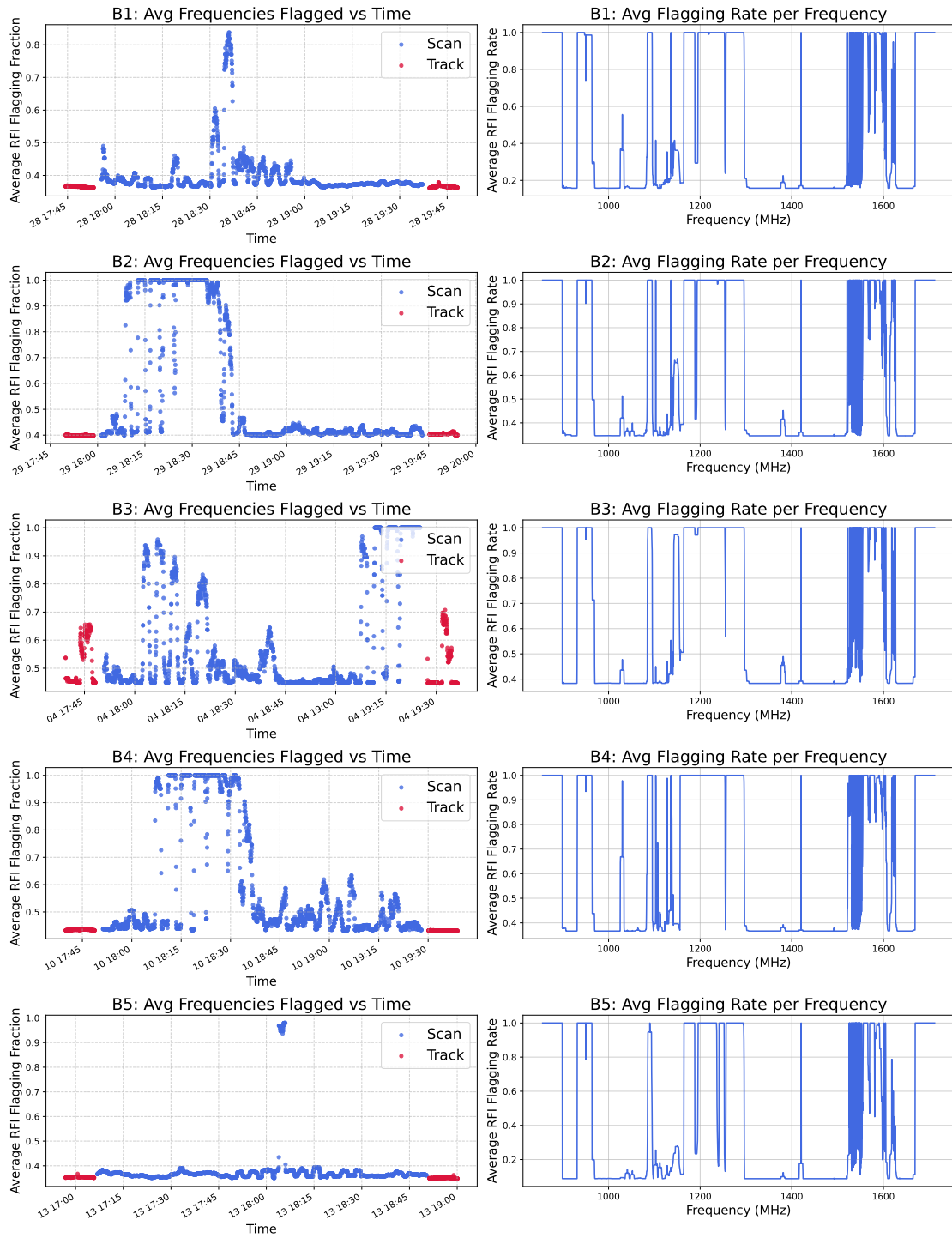


Figure 3.6: Flagging statistics for all five observation blocks. Each row presents two plots per block: the left panel shows the average fraction of frequencies flagged over time, indicating the temporal evolution of RFI contamination, while the right panel displays the average flagging rate per frequency channel, highlighting persistent interference at specific frequencies.

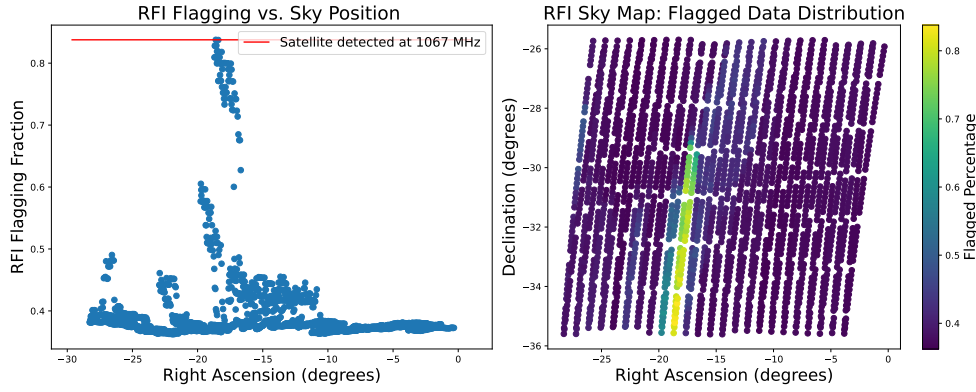


Figure 3.7: Identification of satellite-related RFI in Block 1. The left panel shows the fraction of flagged frequency channels as a function of right ascension (RA), highlighting a peak at 1066.87 MHz indicative of satellite contamination. The right panel presents a sky map of average flagged fractions across RA and declination (DEC), where a localized region of high flagging percentage reveals the spatial coordinates at which the satellite interference is detected.

3.5 Characterisation of per-antenna flagging

Individual antennas may exhibit unique flagging characteristics due to local environmental factors, instrumental effects, or position-dependent interference exposure. To investigate this, the percentage of flagged data is computed for each antenna individually. This calculation is performed separately for both the scan and track modes of each observation block to determine if flagging behavior varies with the observation strategy. The results are visualized using scatter plots that map each antenna to its physical altitude and longitude coordinates.

Figure 3.9 shows the spatial distribution of flagging rates that indicates clear clustering patterns, with antennas at the centre of the array generally exhibiting lower flagging rates than antennas at the periphery of the array.

In addition, persistent antenna-specific anomalies, such as the behaviour of antenna m057 are clearly visible. This type of behavior can be associated with proximity to a stationary RFI source near the specific antenna.

In the next chapter, we will investigate whether the RFI flags of each antenna can be used to infer global geometrical properties of antenna locations. To do so, we will use the Manifold learning technique described in Chapter 2.

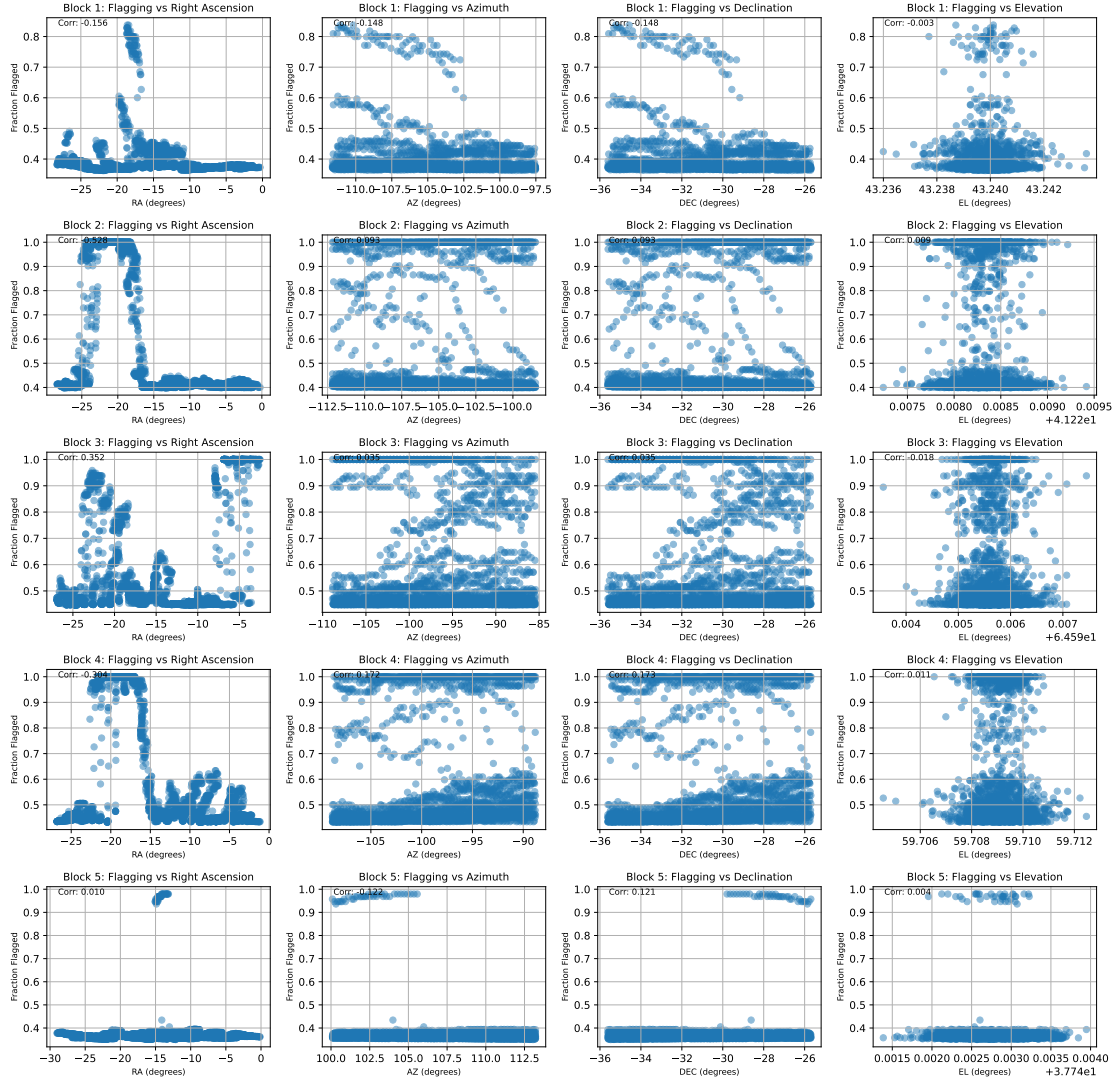


Figure 3.8: Fraction of flagged data as a function of sky coordinates for all five observational blocks during scan time only. Each panel shows the correlation between flagging and one of four pointing parameters: Right Ascension (RA), Azimuth (AZ), Declination (DEC), and Elevation (EL). Notably, Block 2 and Block 3 exhibit stronger correlations with RA, indicating potential directional dependence of RFI.

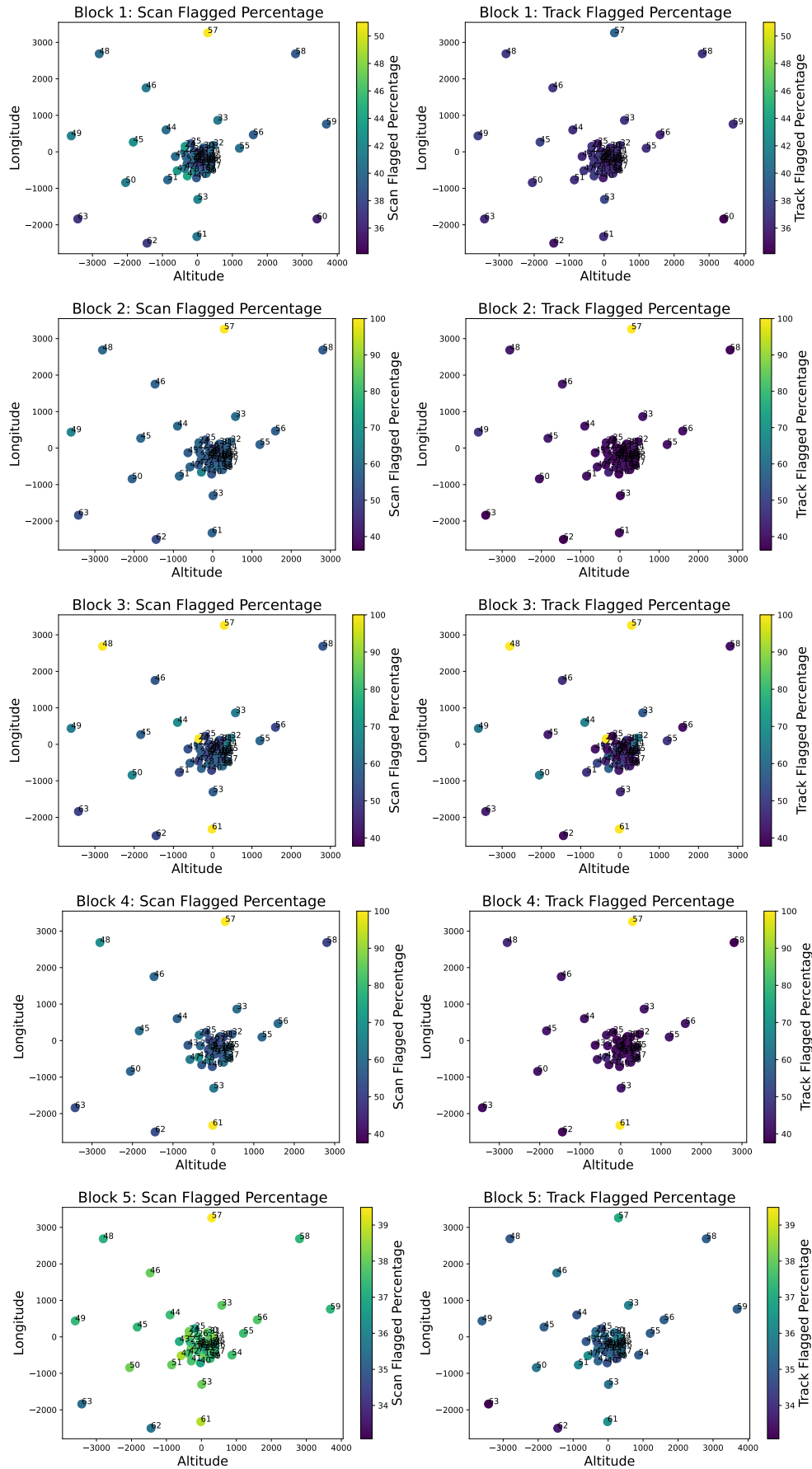


Figure 3.9: Block 1 to 5 flagged percentage scatter

4. Characterising per-antenna flagging

In this chapter, we present the study's core findings and provide an interpretation of what they reveal about RFI patterns in the MeerKLASS data. We start defining a distance between per-antenna flags using the Hamming distance. We then use these distances in the context of manifold learning: we test if it is possible to reveal a meaningful two-dimensional structure that is embedded within our high-dimensional data. In particular, we are interested in recognizing correlation with the physical positions of the antennas that should be more present the more local origin of the RFI is. In other words, we would like to test if the RFI flagging statistics inherently contain information about the physical geometry of the telescope array.

4.1 Distance Matrices

Physical Distance Matrix

The physical distance matrix establishes the ground truth spatial relationships between the antennas and serves as the reference for evaluating spatial correlations in the RFI patterns.

For each pair of antennas i and j with coordinates $(\text{alt}_i, \text{lon}_i)$ and $(\text{alt}_j, \text{lon}_j)$, the 2D Euclidean distance in the ground plane is computed as follows:

$$d_{\text{phys}}(i, j) = \sqrt{(\text{alt}_i - \text{alt}_j)^2 + (\text{lon}_i - \text{lon}_j)^2} \quad (4.1.1)$$

To facilitate comparison with other distance-based similarity metrics, the physical distances are normalized by the maximum antenna distance observed in the array:

$$d_{\text{norm}}(i, j) = \frac{d_{\text{phys}}(i, j)}{\max d_{\text{phys}}} \quad (4.1.2)$$

This normalization leads to a symmetrical matrix of distances in the range $[0, 1]$, where 0 corresponds to the same antenna pair and 1 represents the most distant antenna pair in the array.

Hamming distance matrix

The Hamming distance matrix quantifies the dissimilarity in the RFI flagging patterns between pairs of antennas, potentially encoding spatial correlation information.

The Hamming distance, originally developed for error detection in digital communication, measures the number of positions where two binary sequences differ. We adapt this concept to quantify the differences in RFI flagging patterns between pairs of antennas.

For antennas i and j , we compute the raw Hamming distance over all time–frequency cells as:

$$H_{\text{raw}}(i, j) = \sum_{t=1}^T \sum_{f=1}^F (\text{flags}_{i,t,f} \oplus \text{flags}_{j,t,f}), \quad (4.1.3)$$

where $\oplus$ denotes the logical XOR operation that results in 1 if the flags differ and 0 if they are identical.

To enable a consistent comparison between datasets with different durations or resolutions, the raw Hamming distances are normalised:

$$H(i, j) = \frac{H_{\text{raw}}(i, j)}{T \cdot F}, \quad (4.1.4)$$

where T and F are the total number of time steps and frequency channels, respectively. The resulting normalised Hamming distance $H(i, j)$ lies in the interval $[0, 1]$, where:

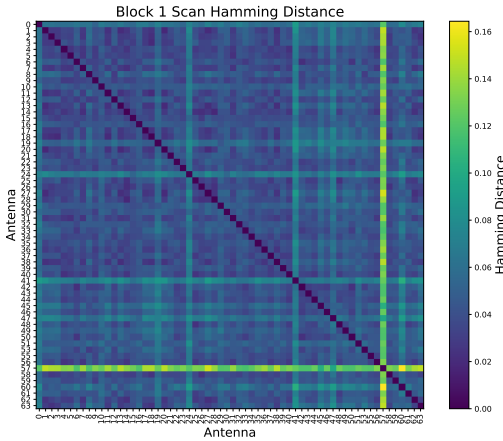
- $H = 0$: Identical flagging patterns (perfect similarity),
- $H = 1$: Completely opposite flagging patterns (maximum dissimilarity),

The process of calculating the normalized Hamming distance, $H(i, j)$, is repeated for all unique pairs of antennas (i, j) in the array. These pairwise dissimilarity values are then compiled into a comprehensive $n \times n$ matrix, where n is the number of antennas.

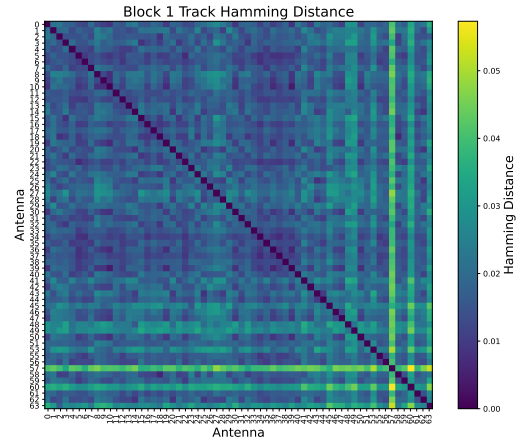
This final matrix is referred to as the *Hamming distance matrix*, denoted as D_H . By construction, the matrix D_H is symmetric, satisfying the property: $D_H(i, j) = D_H(j, i)$ and all diagonal elements are zero, as: $D_H(i, i) = 0$

We compute separate Hamming distance matrices for different observation modes and blocks to allow comparison between different observation periods and to assess whether spatial correlations vary according to observation strategy. We present the results in figures from 4.1 to 4.5. From these figures, we can infer that:

- Antennas that are close often exhibit similar flagging behavior.
- Certain antennas show consistently distinct flagging patterns regardless of the observation period.
- The overall structure of similarity in flagging behavior is relatively stable between observation blocks.

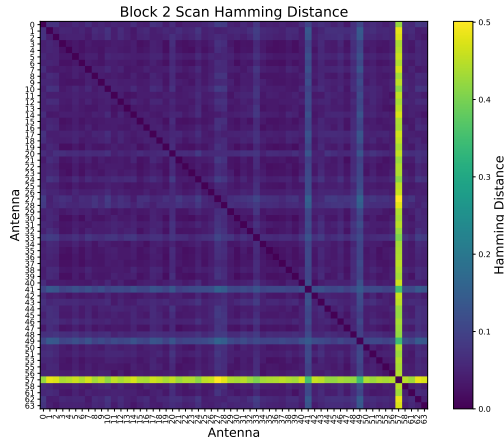


(a) Scan Hamming distance matrix

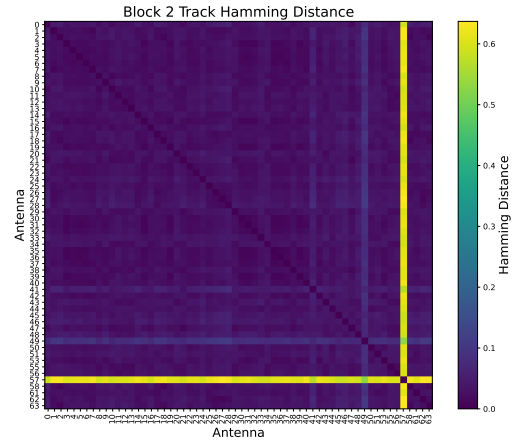


(b) Track Hamming distance matrix.

Figure 4.1: Hamming distance matrices for the RFI flagging patterns of Block 1, separated by observational mode. The color of each cell indicates the dissimilarity between a pair of antennas, with brighter colors representing a greater difference in their flagging patterns.

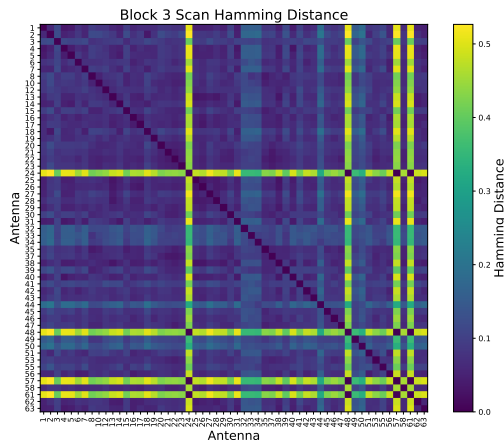


(a) Scan Hamming distance matrix

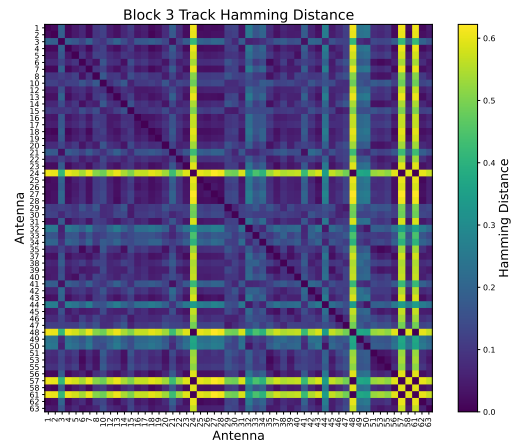


(b) Track Hamming distance matrix

Figure 4.2: Pairwise dissimilarity matrices for Block 2, based on the Hamming distance between antenna RFI flags. Brighter colors indicate a higher degree of dissimilarity between the flagging patterns of antenna pairs.



(a) Scan Hamming distance matrix



(b) Track Hamming distance matrix

Figure 4.3: Pairwise dissimilarity matrices for Block 3, based on the Hamming distance between antenna RFI flags. Brighter colors indicate a higher degree of dissimilarity between the flagging patterns of antenna pairs.

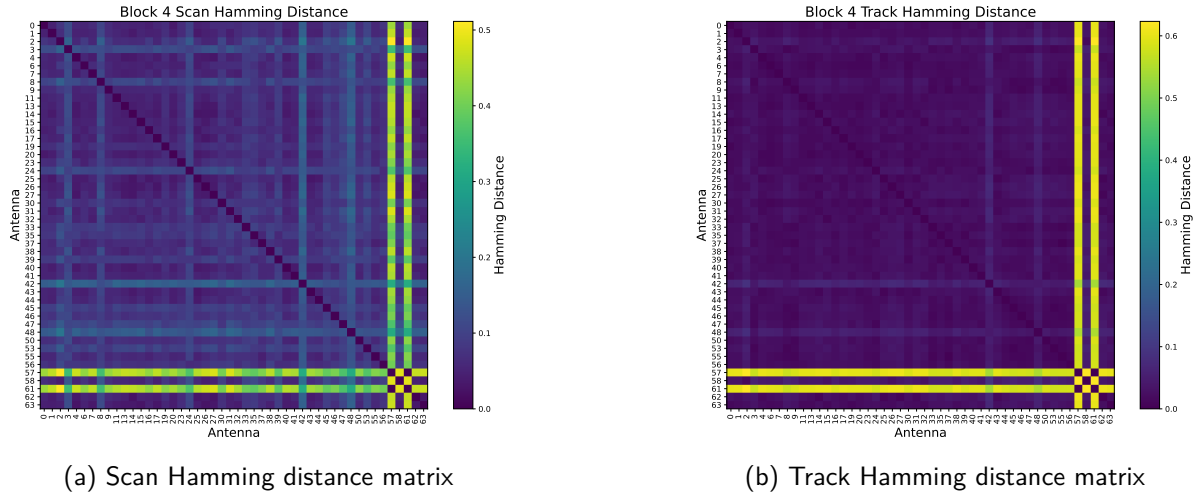


Figure 4.4: Pairwise dissimilarity matrices for Block 4, based on the Hamming distance between antenna RFI flags. Brighter colors indicate a higher degree of dissimilarity between the flagging patterns of antenna pairs.

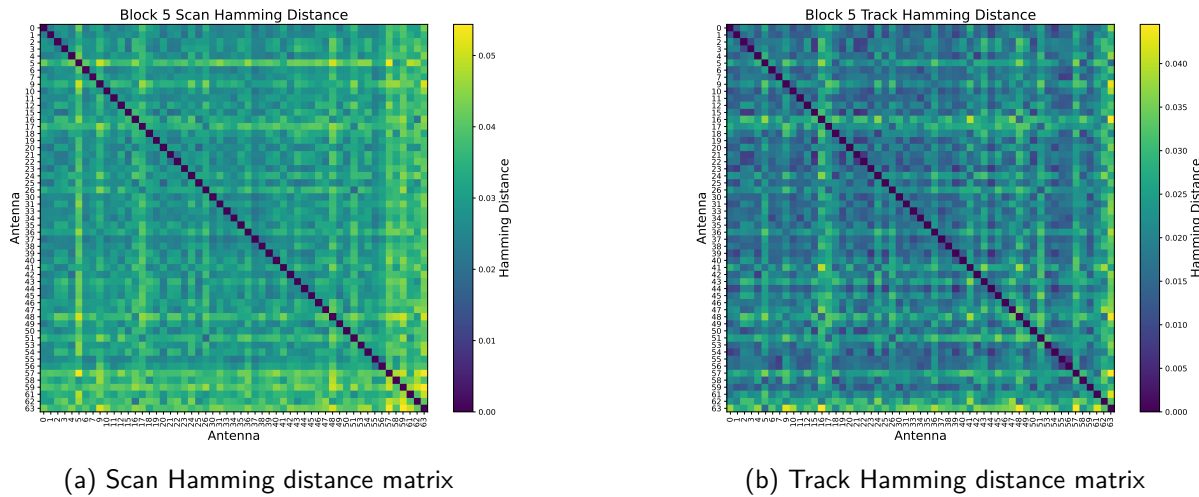


Figure 4.5: Pairwise dissimilarity matrices for Block 3, based on the Hamming distance between antenna RFI flags. Brighter colors indicate a higher degree of dissimilarity between the flagging patterns of antenna pairs.

Block	Mode	Pearson Correlation	Spearman Correlation
B1 (1640712986)	Track	$r = 0.5221$	$\rho = 0.4174$
B5 (1631552188)	Scan	$r = 0.4931$	$\rho = 0.4159$
B2 (1640799689)	Scan	$r = 0.3815$	$\rho = 0.3232$
B3 (1638639082)	Scan	$r = 0.3774$	$\rho = 0.3191$
B2 (1640799689)	Track	$r = 0.3633$	$\rho = 0.4107$
B3 (1638639082)	Track	$r = 0.3639$	$\rho = 0.2716$
B4 (1639157507)	Scan	$r = 0.3564$	$\rho = 0.3026$
B4 (1639157507)	Track	$r = 0.3491$	$\rho = 0.3529$
B1 (1640712986)	Scan	$r = 0.2746$	$\rho = 0.2305$
B5 (1631552188)	Track	$r = 0.2813$	$\rho = 0.2332$

Table 4.1: Correlation between physical antenna separation and Hamming distance of RFI flagging patterns for scan and track modes across all blocks.

4.2 Correlation Analysis

The first test that can be done is to check how the physical distance correlates with the result from the Hamming distance. The correlation analysis between the Hamming and physical distance matrices forms the empirical basis for testing our central hypothesis about the spatial organization of RFI patterns.

Hypothesis: *The spatial proximity hypothesis states that antennas in close spatial proximity will have similar RFI exposure patterns.* If this holds, we expect:

Physically close antennas $\rightarrow$ similar RFI patterns $\rightarrow$ small Hamming distances,
 Physically distant antennas $\rightarrow$ different RFI patterns $\rightarrow$ large Hamming distances.

This would manifest as a positive correlation between the physical and Hamming distance matrices.

We computed the correlations between the distance matrices using a vectorised correlation analysis approach. Specifically, the upper triangular parts (excluding the diagonal elements) of both the Hamming and physical distance matrices for the five blocks are extracted and flattened into 1D vectors. Each element in these vectors represents a unique antenna pair, thus enabling standard statistical correlation techniques.

Two correlation coefficients are computed:

- **Pearson correlation:** Measures the strength of a linear relationship between physical and Hamming distances.
- **Spearman correlation:** Captures monotonic relationships (increasing or decreasing), which may not be strictly linear. It is less sensitive to outliers and more robust to nonlinear transformations.

Table 4.1 shows the results for the correlation analysis done on the 5 observation blocks between the physical distance and the hamming distance matrices of the antennas.

The findings indicate that there is some correlation but that the RFI structure has a large non-local component. There is, however, potential value in employing spatial information in RFI mitigation strategies, whether in array design or observation scheduling.

4.3 Multidimensional Scaling

4.3.1 Multidimensional Scaling (MDS): Theoretical Foundation

Multidimensional Scaling (MDS) refers to a family of dimensionality reduction techniques that aim to embed high-dimensional dissimilarity data into lower-dimensional Euclidean spaces while preserving the pairwise distance structure. In the context of this study, MDS is applied to Hamming distance matrices to generate interpretable 2D representations that may reveal underlying spatial organization patterns among antennas.

4.3.2 Classical MDS Formulation

Let δ_{ij} denote the dissimilarity (e.g., Hamming distance) between antennas i and j . Classical MDS aims to find a configuration of points $Y = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\} \subset \mathbb{R}^d$ such that the Euclidean distances between embedded points $d_{ij}(Y) = \|\mathbf{y}_i - \mathbf{y}_j\|$ approximate δ_{ij} . The optimisation objective is:

$$\text{STRESS}(Y) = \sum_{i=1}^n \sum_{j=1}^n (\delta_{ij} - d_{ij}(Y))^2 \quad (4.3.1)$$

The goal is to minimise the STRESS value so that the embedded configuration best preserves the structure of the input dissimilarities.

4.3.3 MDS Calculation

We apply the MDS algorithm to the $n \times n$ Hamming distance matrix, D_H , where n is the number of active antennas and each element $\delta_{ij} \in D_H$ represents the dissimilarity between the RFI flagging patterns of antennas i and j . The objective is to find a configuration of n points in a low-dimensional space, $\mathcal{Y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ where $\mathbf{y}_i \in \mathbb{R}^d$, that best reproduces the distances in D_H .

For this study, the target dimensionality is set to $d = 2$. The algorithm iteratively adjusts the positions of the points $\mathbf{y}_i$ in the 2D plane to minimize the *STRESS* function defined in Equation 4.3.1.

The final output of this procedure is a coordinate matrix $Y \in \mathbb{R}^{n \times 2}$, where each row contains the derived x and y coordinates of an antenna in the new embedded space. This calculation is performed independently for the Hamming distance matrices derived from both the scan and track modes of each observation block.

To quantitatively evaluate the quality of the 2D MDS embeddings, we analyse how much the physical structure of the array is preserved. For this purpose, the nearest neighbours of each antenna at its original physical location are compared with its nearest neighbours in the new embedded configuration.

Let $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ be the set of physical coordinates for the n antennas, where $\mathbf{p}_i \in \mathbb{R}^2$, and let $\mathcal{Y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ be the set of coordinates derived from the MDS embedding. For each antenna i , we define its k -nearest neighbourhood in physical space, $\mathcal{N}_{\mathcal{P}}(i, k)$, as the set of k antennas that are closest to $\mathbf{p}_i$. Similarly, we define the neighbourhood in the embedded space, $\mathcal{N}_{\mathcal{E}}(i, k)$, as the set of k antennas closest to $\mathbf{y}_i$.

The accuracy of the embedding for antenna i is quantified by a percentage of mismatch in the neighbourhood, which is defined as follows:

$$\text{Mismatch}\%(i) = \frac{k - |\mathcal{N}_{\mathcal{P}}(i, k) \cap \mathcal{N}_{\mathcal{E}}(i, k)|}{k} \times 100\% \quad (4.3.2)$$

where $|\mathcal{N}_{\mathcal{P}}(i, k) \cap \mathcal{N}_{\mathcal{E}}(i, k)|$ is the number of common antennas in both neighbourhood sets. A value of 0% indicates perfect preservation of the local neighbourhood, while 100% indicates complete divergence.

The first set of results, shown in Figures 4.6 – 4.10, shows the MDS embeddings using the full, unfiltered Hamming distance matrices for all five observation blocks with antennas colour-coded according to the neighborhood preservation quality measure.

We refer instead to Appendix A for an alternative, more simplistic, colour coding scheme where each point is coloured according to the antenna number.

Block-specific performance:

- **Block 1:** The percentage difference between the physical and embedded neighborhoods ranges from 50% to 100%, indicating moderate spatial preservation. Notably, antenna 57 exhibits poor neighborhood retention as it consistently shows up as a spatial outlier in both scan and track modes.
- **Block 2:** This block exhibits good spatial coherence, especially in the scan mode. The majority of antennas exhibit 50–80% neighborhood preservation, and the embedding's low variability and central clustering indicate that physical spatial relationships have been moderately recovered.
- **Block 3:** This block Shows clear distinctions between track and scan modes. The track embedding is dominated by closely clustered points and multiple distinct outliers, such as antennas 24, 48, 57 and 61, whereas the scan embedding displays a more uniform spatial distribution.
- **Block 4:** This block reveals an embedding that is central. However, antennas 57 and 61 exhibits systematic spatial deviation once more, confirming its status as a persistent anomaly.
- **Block 5:** This block achieves the most consistent and stable spatial arrangement in both modes of observation. The percentage differences range from 50% to 90%, and the embeddings are closely clustered, indicating little distortion during dimensionality reduction.

Mode-dependent variations (scan vs. track):

A consistent pattern observed in all blocks is the structural difference between scan and track mode embeddings. Scan mode embeddings tend to be more spatially distributed and better reflect the physical arrangement of the antennas in cases such as blocks 2 and 5. This is likely due to the fact that during a long scan across a relatively quiet sky, the dominant sources of correlated RFI are external (e.g. ground-based transmitters or distant satellites). The effect of these signals is primarily modulated by the physical distance and orientation of the antennas, so the Hamming distance is a good indicator of physical distance.

Tracking data are consistently more compact, with antennas forming tighter clusters. This is to be expected as all antennas are pointed at a single, bright calibration source during tracking. This strong common mode signal can dominate the flagging statistics for all antennas, making them appear more similar to each other. This reduces the overall spread of the Hamming distances and causes the MDS algorithm to place them closer together in the embedded space.

The most significant finding from this initial analysis is that the embeddings consistently reveal non-random spatial patterns. The fact that the antennas cluster in a recognisable way and do not appear

as a random scatter is the primary evidence for the hypothesis of this study. It confirms that the dissimilarities in flagging patterns between the antennas are not random, but are correlated in a way that the MDS algorithm can interpret as geometric structure.

Additionally, in all blocks, antenna 57 proves to be a recurring outlier, often positioned at the edge of the embedded space and showing neighborhood differences of more than 90%. This repeated anomaly suggests a possible hardware or environmental issue specific to this antenna, consistent with what we know about this antenna and its relatively close position to the nearest cellphone tower to the MeerKAT telescope.

As a consequence, this initial embedding may be sensitive to distortions caused by antennas with highly anomalous flagging patterns, necessitating subsequent refinement of the analysis. This is discussed further in the next section.

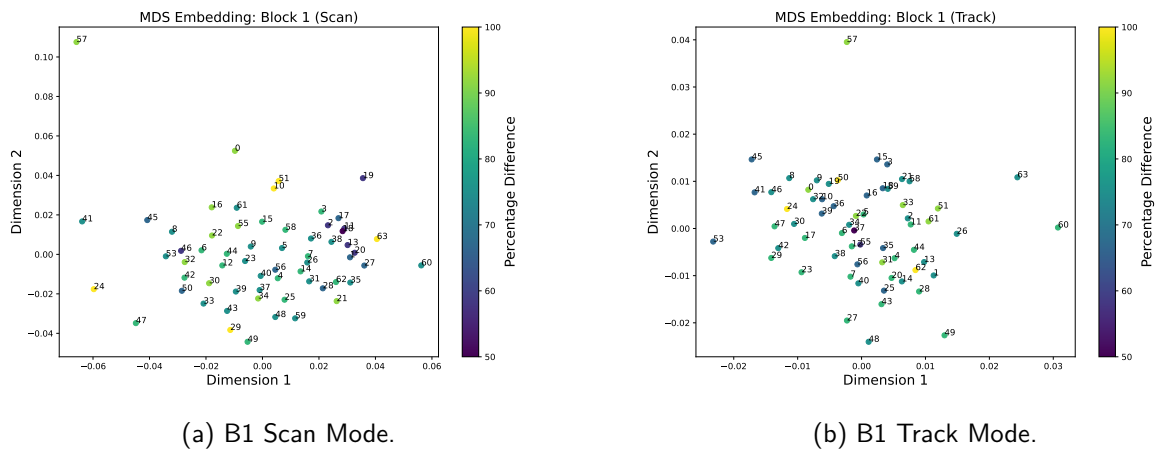


Figure 4.6: Comparison of MDS embeddings for Block 1 (B1) during scan and track modes. Lower percentage values indicate better correspondence with physical antenna layout.

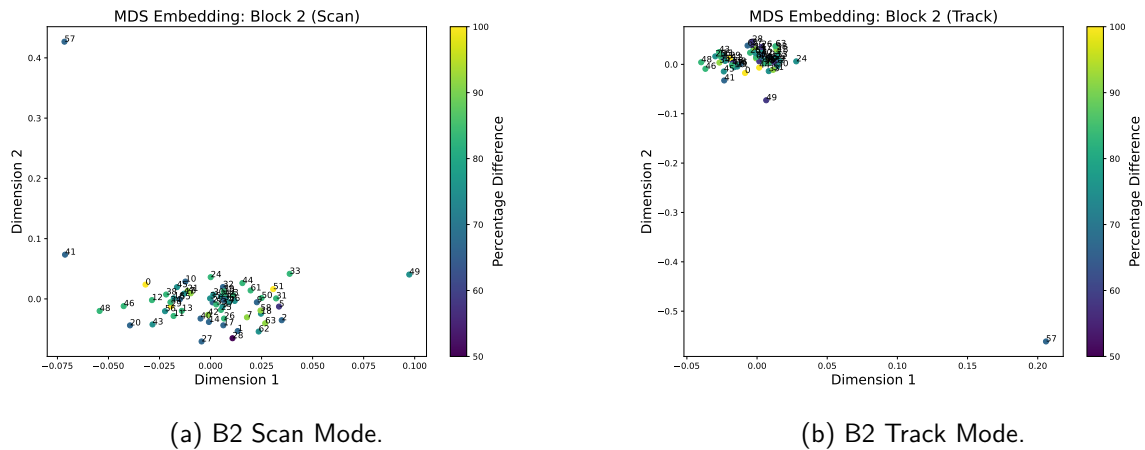


Figure 4.7: Comparison of MDS embeddings for Block 2 (B2) during scan and track modes. Lower percentage values indicate better correspondence with physical antenna layout.

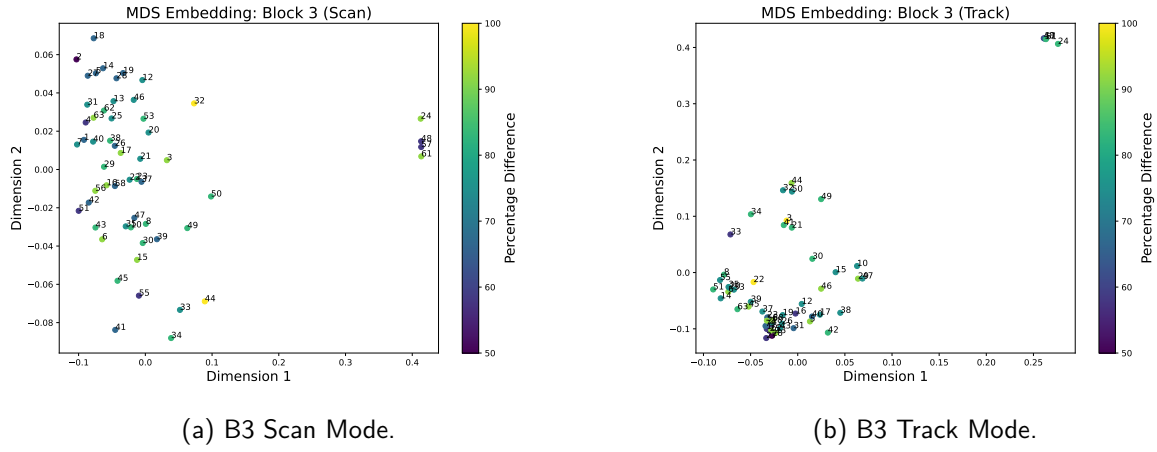


Figure 4.8: Comparison of MDS embeddings for Block 3 (B3) during scan and track modes. Lower percentage values indicate better correspondence with physical antenna layout.

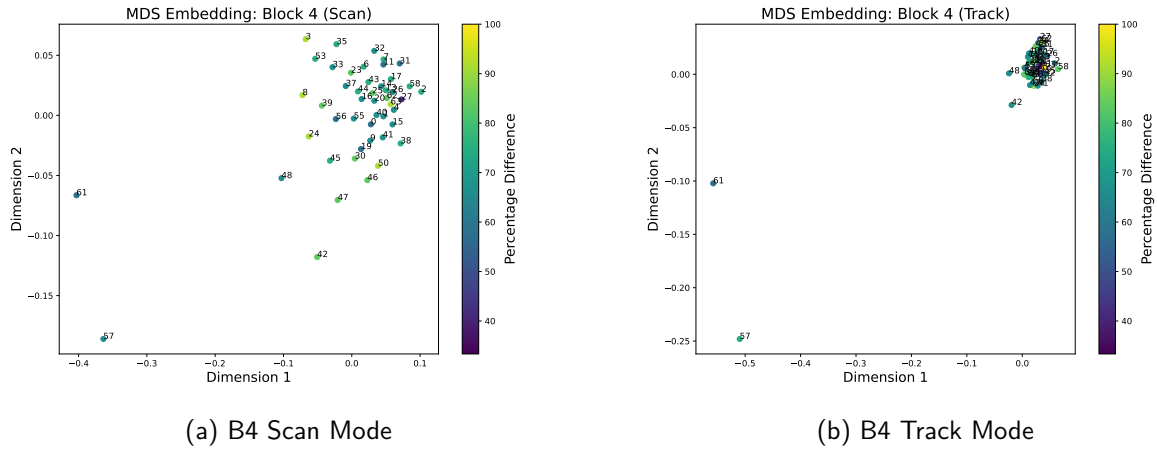


Figure 4.9: Comparison of MDS embeddings for Block 4 (B4) during scan and track modes. Lower percentage values indicate better correspondence with physical antenna layout.

4.3.4 Outlier Removal and Refined MDS Calculation

Antennas with systematically anomalous RFI flagging behaviour can act as outliers, disproportionately affecting the MDS algorithm and potentially distorting the resulting spatial embedding. To mitigate this effect, a method for detecting and removing outliers is applied to the Hamming distance matrices before a second, refined MDS calculation is performed.

The method identifies outliers based on the interquartile range (IQR) of the mean Hamming distances. For each antenna i , its mean Hamming distance, $\bar{\delta}_i$, is calculated by averaging its dissimilarity to all other antennas in the array:

$$\bar{\delta}_i = \frac{1}{n-1} \sum_{j \neq i} \delta_{ij} \quad (4.3.3)$$

This metric, $\bar{\delta}_i$, indicates how much the RFI pattern of an antenna differs on average from the rest of

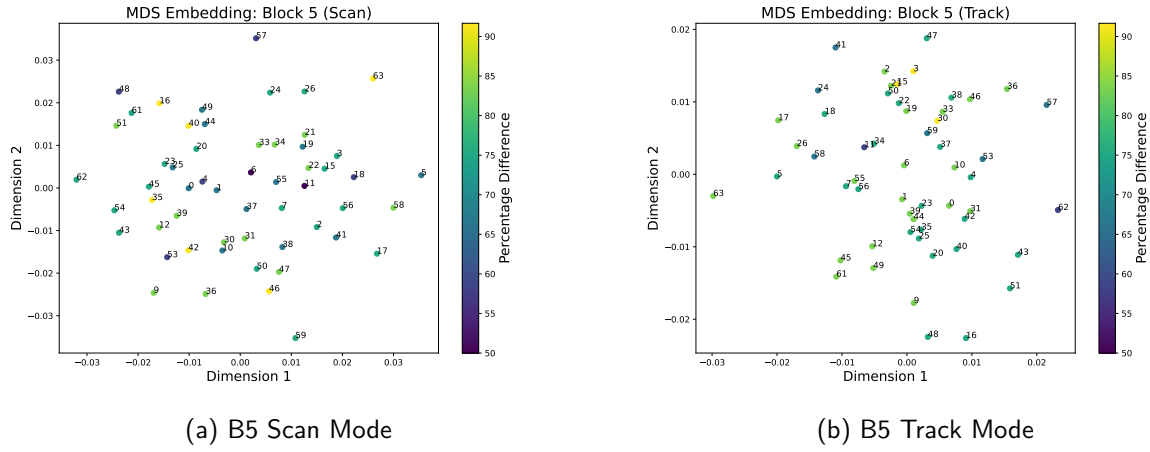


Figure 4.10: Comparison of MDS embeddings for Block 5 (B5) during scan and track modes. Lower percentage values indicate better correspondence with physical antenna layout.

the array. Outliers are then identified based on the distribution of these mean distances. We define the first quartile (Q_1) and third quartile (Q_3) of the set of all $\bar{\delta}_i$ values, and the interquartile range as $IQR = Q_3 - Q_1$. An antenna is flagged as an outlier if its mean Hamming distance $\bar{\delta}_i$ falls outside the range defined by:

$$[Q_1 - c \cdot IQR, \quad Q_3 + c \cdot IQR] \quad (4.3.4)$$

where c is a constant multiplier, typically set to 1.5.

Once the outlier antennas are identified, they are removed by deleting the corresponding rows and columns from the Hamming distance matrix. The MDS algorithm, as described before (Section 4.3.3), is then reapplied to this cleaned and reduced matrix to produce a refined 2D embedding. This process is performed for both scan and track modes, and the results are presented below in Figures 4.11 to 4.14.

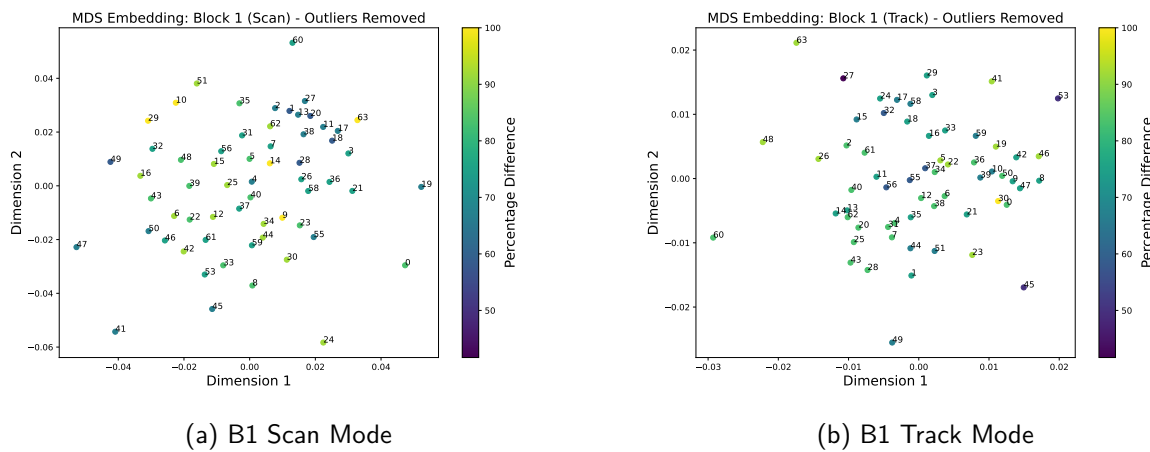


Figure 4.11: MDS embeddings for Block 1 after outlier removal. The removal enhances the visibility of underlying structure, particularly in the track mode which becomes more compact.

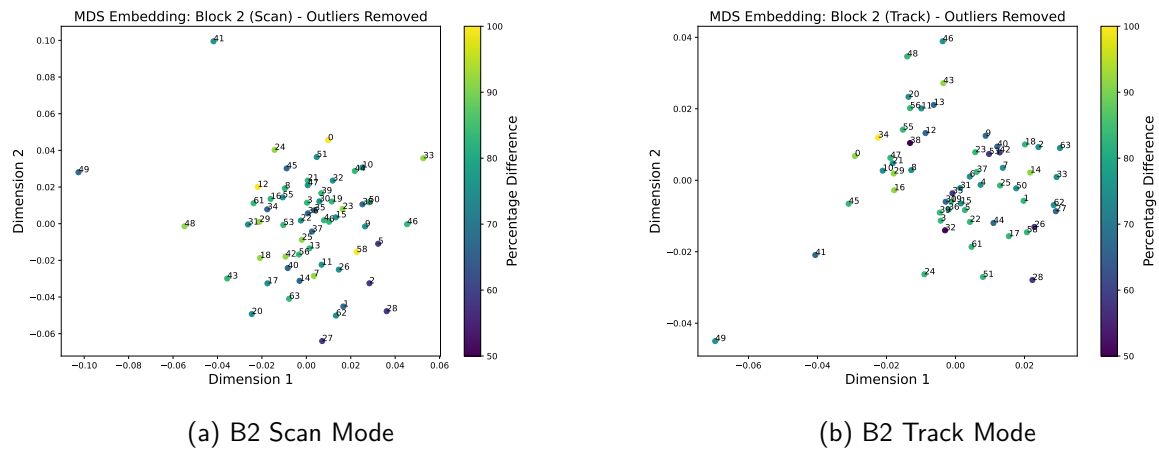


Figure 4.12: MDS embeddings for Block 2 after removing high-dissimilarity antennas. Scan mode becomes more geometrically interpretable, while track mode shows a well-consolidated cluster.

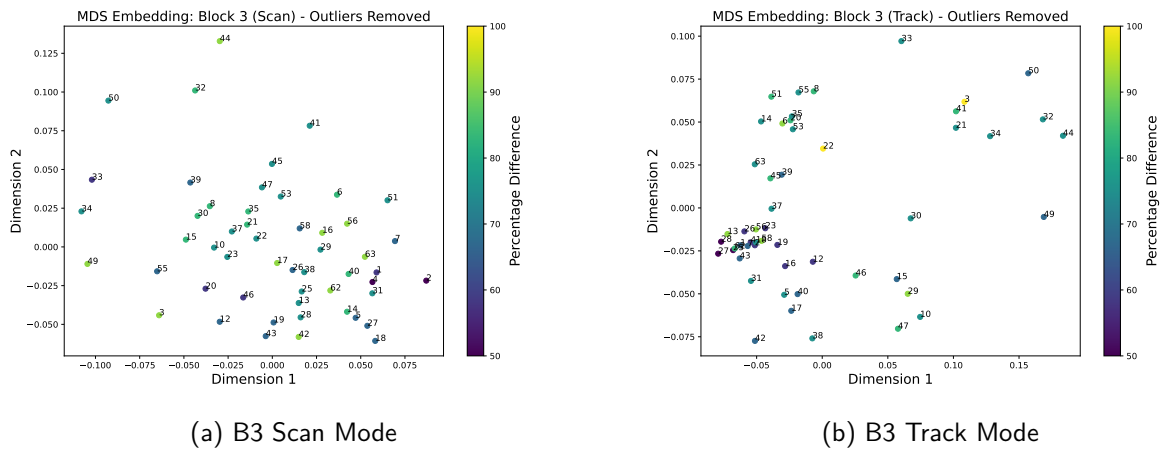


Figure 4.13: Cleaned Block 3 embeddings illustrating improved spatial coherence in both modes. Track mode particularly benefits from reduced peripheral outliers.

Block-specific performance:

- **Block 1:** The scan mode embedding (Figure 4.11a) becomes more evenly distributed after outliers are eliminated, with antennas arranged in a more clear 2D spatial layout and neighbourhood preservation rising to the 50–90% range. A compact central cluster with better symmetry and less external distortion can be seen in the track mode (Figure 4.11b).
- **Block 2:** Eliminating outliers has a significant positive impact on both modes. With well-separated antenna groups and preservation increasing to 50–85%, the scan mode embedding (Figure 4.12a) changes from scattered to structured. With the majority of antennas forming a central geometric cluster and few dissimilarity outliers, the track mode (Figure 4.12b) exhibits even tighter spatial cohesion.
- **Block 3:** After removing problematic antennas, the cleaned scan mode embedding (Figure 4.13a) shows a more balanced spatial configuration, greatly lowering noise from edge points. After

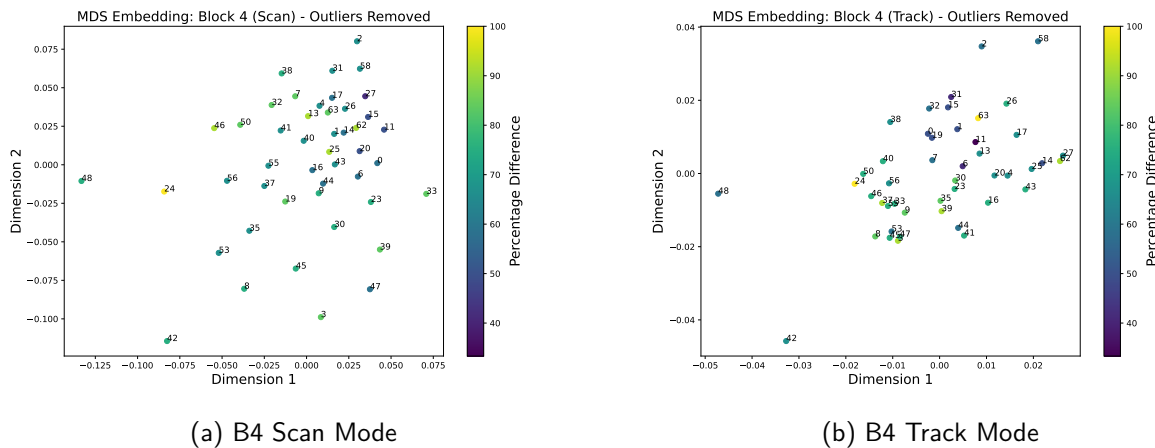


Figure 4.14: Block 4 embeddings post outlier filtering, with improved concentration and reduced fragmentation in antenna placement across both scan and track views.

removing extreme outliers, particularly antennas 24 and 57, the track mode (Figure 4.13b) reveals an orderly cluster with strong neighbourhood overlap (usually 50–80%).

- **Block 4:** The cleaned embeddings in scan mode (Figure 4.14a) and track mode (Figure 4.14b) show improved spatial regularity and reduced scatter. The removal of outliers improves interpretability as the majority of the antennas show tighter geometric grouping and solid preservation.
- **Block 5:** No antennas were flagged as outliers based on the percentile and IQR thresholds. Therefore, Block 5 embeddings remained unchanged from Section 4.3.3, already demonstrating strong spatial preservation across both scan and track modes.

Mode-dependent variations (scan vs. track):

Even after removing the outlier antennas, the fundamental structural differences between scan and track modes remain in the refined embeddings. The track mode embeddings consistently result in more compact and tightly grouped clusters (e.g. Figure 4.11b, 4.12b), while the scan mode embeddings remain more spatially dispersed and show a clearer correspondence with the physical antenna layout. This consistency suggests that the tight clustering in track mode is not an artefact of outlier antennas, but rather an inherent feature of the observation strategy. The strong common mode signal from the bright calibration source during tracking makes all antennas appear more similar in their flagging behaviour, an effect that persists even after excluding the most anomalous antennas.

Notably, Block 5 required no outlier removal in order to achieve a stable and spatially coherent embedding. This block can be interpreted as a “best-case scenario” and probably represents an observation in which the RFI environment was more uniform throughout the array and no single antenna was dominated by a strong, local interference source.

These results show that removing a small number of outlier antennas systematically improves the spatial coherence of the embeddings, leading to a clearer recovery of the physical array configuration. This finding supports the concept that the RFI signal consists of at least two components: a widespread, geometrically correlated pattern that is visible to most of the array, and a localised, antenna-specific pattern that affects only a few antennas. By removing the antennas that are dominated by the localised RFI, the underlying geometric structure shared by the majority of the array becomes much clearer.

5. Conclusion and Future Work

This thesis presented a novel approach for characterizing radio frequency interference (RFI) in single-dish observations with the MeerKAT telescope (the MeerKLASS survey data), in which RFI flagging information is studied and compared to the spatial configuration of the instrument to improve our understanding of the systematics. The main objective was to determine whether RFI flagging patterns across the MeerKAT array contain information about the physical disposition of the antennas. This study suggests that such a relationship exists and can be uncovered using a combination of dissimilarity metrics and manifold learning.

Across all five observation blocks, a positive correlation was found between the Hamming distance of the RFI flag masks and the actual physical separation of the antennas (Pearson $r \approx 0.28$ – 0.52 ; Spearman $\rho \approx 0.23$ – 0.42). This is evidence that the antennas that experience similar interference tend to be those that are physically close. When this dissimilarity was used to create a two-dimensional map with Multidimensional Scaling, the resulting embeddings consistently showed a non-random structure that roughly matched the known array layout. Although the initial reconstruction is imperfect and neighbourhood divergence rates are typically in the range of 50–70 %, the methodology presented allows to improve our knowledge on the data.

Furthermore, the study showed a clear way to improve this reconstruction. After the statistical removal of outlier antennas, most notably antenna m057 (and occasionally antenna m061), which was consistently identified in every block, the refined MDS embedding showed improved accuracy and a visibly closer match to the actual array structure. This not only confirms the diagnostic value of the method for detecting antennas with unique hardware or environmental issues, but also proves that the geometric signal can be enhanced by addressing localised RFI effects.

While the results do not yet represent a perfect reconstruction of the array, they successfully validate this method as a promising new diagnostic tool. The ability to extract geometric information from RFI flags alone opens a new perspective on RFI analysis and emphasises its potential to serve as a valuable probe for the instrument and its environment.

Future work will involve the expansion of this framework by investigating a range of dissimilarity measures and additional manifold learning methods for improving analysis. The portability of the present methodology facilitates its application to radio interferometers like the SKAO, enabling comparative RFI environment studies. Furthermore, we see considerable potential for integration of this analysis with other instrumental diagnostic methods and for application in automated flagging systems, thereby making it an effective tool for improving data quality that are key to solve major open scientific questions.

Acknowledgements

I would like to express my gratitude to Dr. Marta Spinelli, my supervisor, for her invaluable advice, constant support, and guidance throughout this research project. I would also like to thank my family and friends for their unwavering support and faith in my capabilities. Finally, I would like to thank AIMS for providing a unique opportunity and stimulating environment for me to develop as a researcher.

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AppendixA. Appendix

This appendix contains additional visualisations for the refined MDS embeddings presented in the main text. The following figures (A.1 to A.4) show the two-dimensional spatial configurations for observation blocks 1 to 4 after applying the outlier removal procedure. While the embeddings themselves are identical to those analysed in Section 4.3.4, these plots use an alternative colour coding scheme. Here, each point representing an antenna is coloured according to its unique index number. This visualisation helps to directly identify specific antennas and shows how antennas with different indices are distributed within the geometric structure recovered from the RFI flagging patterns.

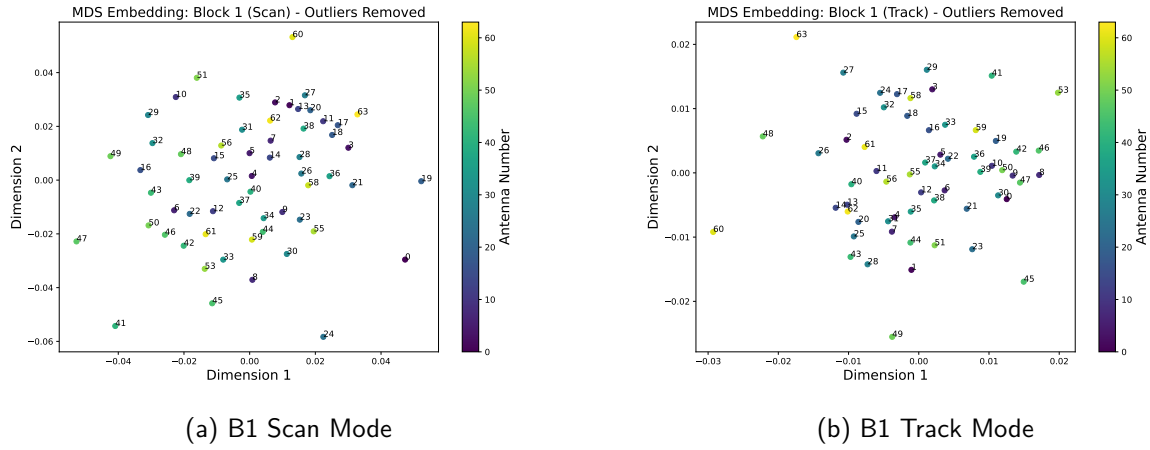


Figure A.1: MDS embeddings for Block 1 after outlier removal. The removal enhances the visibility of underlying structure, particularly in the track mode which becomes more compact.

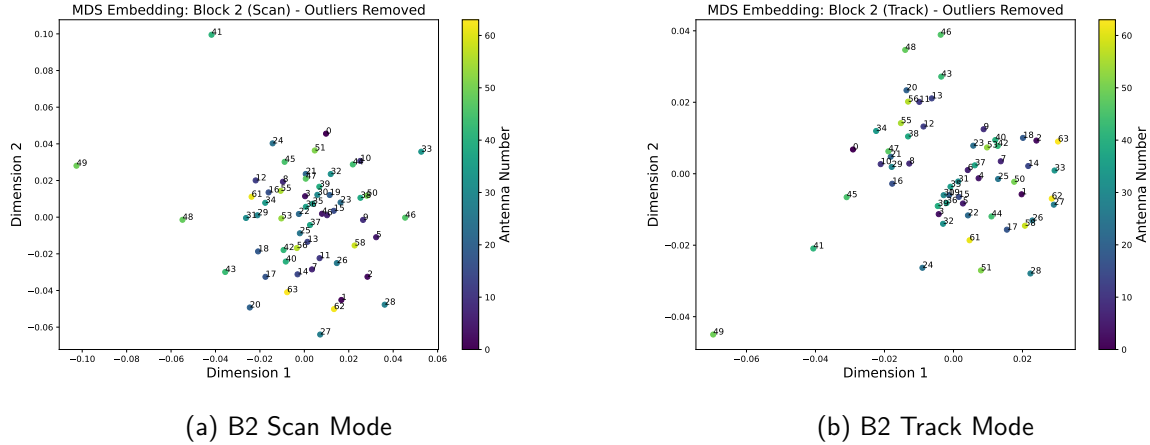
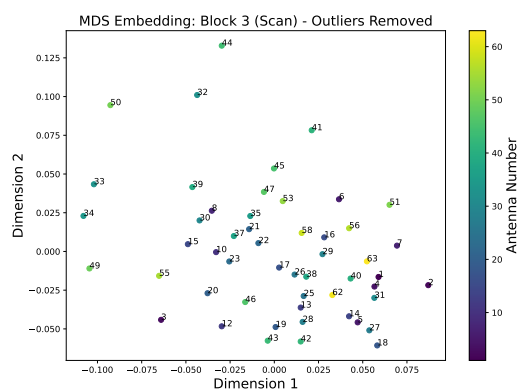
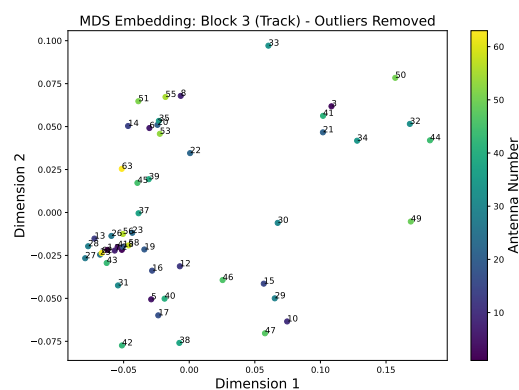


Figure A.2: MDS embeddings for Block 2 after removing high-dissimilarity antennas. Scan mode becomes more geometrically interpretable, while track mode shows a well-consolidated cluster.

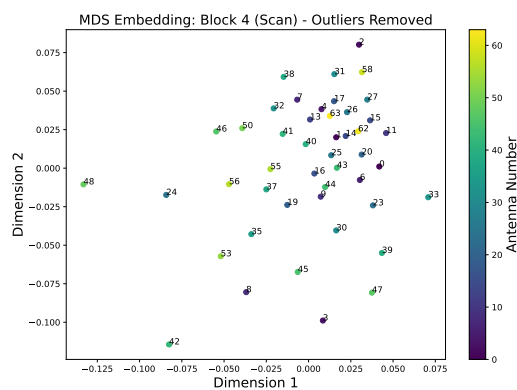


(a) B3 Scan Mode

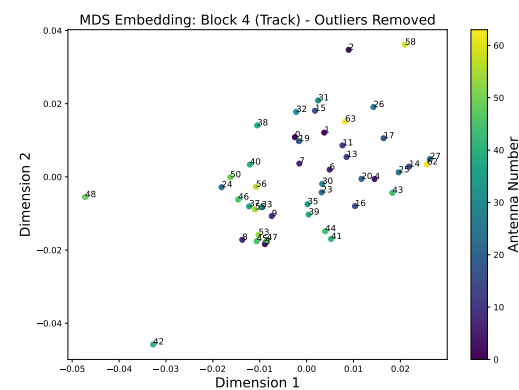


(b) B3 Track Mode

Figure A.3: Cleaned Block 3 embeddings illustrating improved spatial coherence in both modes. Track mode particularly benefits from reduced peripheral outliers.



(a) B4 Scan Mode



(b) B4 Track Mode

Figure A.4: Block 4 embeddings post outlier filtering, with improved concentration and reduced fragmentation in antenna placement across both scan and track views.